SS-L-1682
January 28, 1971
SUPERSEDING
Fed. Spec. SS-C-614
December 23, 1948

FEDERAL SPECIFICATION

LAPPING AND GRINDING COMPOUND, VALVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers two types of valve grinding compound.

1.2 Classification. Valve grinding compounds covered by this specification shall be of the following types, and grits as specified (see 6.2):

Type I - Grease mixed.
Type II - Water mixed.

Grit 100 - Coarse, for chromium-cobalt valve seats.
Grit 120 - Coarse, for fast valve grinding.
Grit 150 - Special medium coarse.
Grit 180 - Medium coarse.
Grit 220 - Medium.
Grit 240 - Fine for chrome-cobalt valve seats.
Grit 280 - Fine, for finish valve grinding, lapping cylinders, and lapping rings into cylinders.
Grit 400 - Extra fine, special purpose for fine lapping, polishing and fitting of parts.
Grit 500 - Extra fine.
Grit 600 - Extra fine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-C-96 - Cans, Metal, 28 Gauge and Lighter.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge form Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 5350

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. Department of Commerce

Grading of Abrasive Grain for Grinding Wheels CS-271.

(Copies of this publication may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Materials. The valve lapping and grinding compound shall be composed essentially of artifical abrasives, and with oil or grease vehicle or with water vehicle. This compound is not affected by exposure at temperatures from-35°F. (37°C.) to 140°F. (60°C.).

3.1.1 Abrasives. Unless otherwise specified, (see 6.2) the abrasive shall be manufactured silicon carbide, manufactured boron carbide, or manufactured aluminum oxide.

3.1.2 Lubricant (Type I only). The vehicle of the Type I compound shall be a good grade of cup grease or mineral oil with lard oil. It shall not be too dry or run too freely.

3.1.3 Vehicle (Type II). The vehicle of water mixed Type II compound shall be starch or such other material which in combination with preservative and water will produce a soft coherent paste, free from "stringiness". The preservative shall be suitable for prevention of mold formation or fermentation when in storage, and from excessive drying when exposed to air.

3.2 Fineness. When the valve grinding compound is tested as specified in 4.4.1 it shall conform to the requirements as expressed in Table I.

3.3 Work valve type I (Grits 120, 150 and 280) only. The Type I valve grinding compound when tested as specified in 4.4.2 in grit sizes Nos. 120, 150, and 280 shall effect a finish characteristic of the grit sizes used, and shall show a minimum work value of 0.225.

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TABLE I. Fineness requirements

Grit No.	No. of sieve thru which 100 percent must pass	Control Sieve		Maximum of oversize on control sieve no.	Minimum thru Control Sieve and retained		Cumulative minimum thru Control Sieve and retained		Maximum of 3 percent thru Sieve No.
		No.	Opening	(Percent)	(Percent)	(Sieve No.)	(Percent)	Sieve No.	
100	70	100	0.0059	20	40	120	65	120 and 170	200
120	80	120	0.0049	20	40	140	65	140 and 170	230
150	100	140	0.0041	15	40	170 and 200	65	170, 200, and 230	325
180	120	170	0.0035	15	40	200 and 230	65	200, 230, and 270	---
220	140	200	0.0029	15	40	230 and 270	60	230, 270 and 325	---
240	170	200	0.0029	5	8	230 and 270	38	230, 270 and 325	---
280	170	230	0.0024	2	12	270 and 325	40	270, 325 and 400	---
400	270	325	0.0017	2	--	---------	--	-------	---
500	325	325	0.0017	--	--	---------	--	-------	---
600	325	325	0.0017	--	--	---------	--	-------	---

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The quality conformance sample shall consist of a sample for tests (4.2.2) and a sample for examination of end item (4.2.3). Samples shall be labeled completely with information identifying the purpose of the sample, name of product, lot number, date of sampling, and contract number.

4.2.1 Lot. All valve-grinding compound manufactured as one batch shall be considered a lot and shall be numbered as such for purposes of inspection.

4.2.2 Sampling for tests. The sample for tests shall consist of two 1 lb. samples taken at random from each lot of valve-grinding compounds. The lots are unacceptable if a sample fails to meet any of the test requirements.

4.2.3 Sampling for examination of end item. A random sample of filled containers and a sample of shipping containers fully prepared for delivery shall be selected from each lot of compound in accordance with MIL-STD-105 at inspection level S2, Acceptance Quality Level (AQL) of 4.0 percent defective, and subjected to examinations specified in 4.3.2.

4.3 Examination.

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4.3.1 End item. The valve-grinding compound shall be examined to determine compliance with requirements not specifically covered by test requirements specified in section 3.

4.3.2 Examination of preparation for delivery. An examination shall be made in accordance with paragraph 60 of PPP-C-96 to determine that cans are closed, packed and marked as required by the appendix to PPP-C-96.

4.4 Test methods.

4.4.1 Fineness.

4.4.1.1 Fineness (100, 120, 150, 180, 220, 240, and 280). The valve-grinding compound in 100, 120, 150, 180, 220, 240, and 280 grit sizes shall be tested for fineness using method described in commercial standard CS-271 (see 2.1), except that limits specified in Table I shall apply (see 3.2).

4.4.1.2 Fineness (400, 500 and 600 grit). The abrasive shall be separated from a 1 pound sample by washing with alcohol or gasoline or other suitable solvent. The abrasive shall be dried and a sample of approximately 10 grams selected by quartering. This sample shall be weighed to the nearest 0.01 gram and placed on a dry No. 325 sieve as specified in CS-271 in a 3 inch diameter frame. A spray nozzle, having seventeen 0.02 inch holes in the tip, shall be connected to a water supply line with a pressure on the spray nozzle of about 10 pounds per square inch and used for washing the sample through the sieve. The washing shall be continued for 5 minutes. The sieve and residue shall be dried in an oven at 110 $\pm$ 10°C, (43 $\pm$ 18°F.) the residue brushed from the sieve, weighed on an analytical balance to the nearest 0.01 gram, and the percentage retained shall be calculated (see 3.2).

4.4.2 Work value type I (grits 120, 150 and 280) only. The test for work value shall be made on a machine designed to simulate the work of grinding an engine poppet value in its seat (see figure 1). The machine shall consist of 3 units with a single driving mechanism. The seat is simulated by a 2 inch diameter, 1/2 inch thick disk of cast iron of 143 Brinell hardness number which shall be superimposed on a steel disk of 267 Brinell hardness number, which simulate a valve. The shaft on which the upper disk is attached shall be rotated by a rack and pinion mechanism which drives it 1-1/4 revolutions in one direction, then a similar number of revolutions in the opposite direction. Once during each such cycle the upper shaft shall be lifted completely from its seat by a cam attached to the rack. The machine shall be driven at a uniform speed of 70 cycles per minute. While seated the disks shall be pressed together with a pressure of 5 pounds per square inch by means of the top weight. Grinding compound shall be fed to the surface between disks through a groove on the seat from graduated compression cups turned by hand every 2 minutes at such a rate as to give 6.5 grams of compound to the surface every hour. Each top and bottom disk shall be cleaned and weight on chemical balances before and after each 2 hour test. A work value for each type shall be determined as follows:

$$\frac{W_1 + W_2}{A} = \text{work value}$$

Where W_1 + weight, in grams ground away from valve.

W_2 = weight, in grams ground away from seat.

A = total weight of compound used, in grams.

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1. Weights equivalent to 5 lbs. per square inch on the disks.
2. Reciprocating rack.
3. Weighed abrasion disk — cast iron, Brinell 143.
4. Weighed abraison disk — steel, Brinell 267.
5. Driving wheel.
6. Idler guide.
7. Crank.
8. Cycle counter.
9. Eccentric.
10. Graduated measuring cup for valve grinding compound. Speed 70 reversals per minute.

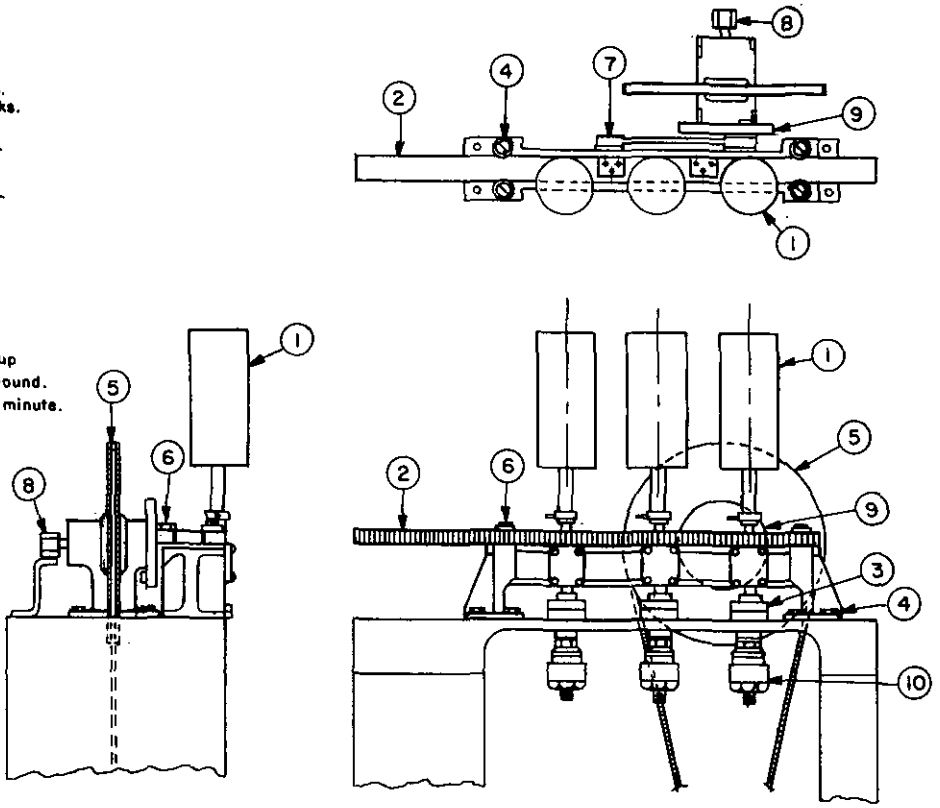


Figure 1. Valve-grinding compound testing machine.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A and B. Valve grinding compound in the quantity specified shall be packaged in a close fitting can conforming to PPP-C-96, type V, class 2.

5.1.2 Level C. The lapping and polishing compound shall packaged to afford the minimum degree of protection necessary to prevent damage during shipment under environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2), in accordance with the appendix to PPP-C-96.

6. NOTE

6.1 Intended use. Grinding compound of grit numbers 120, 150, 220 and 280 covered by this specification is intended for use in valve grinding, and is generally designated as coarse, medium coarse, medium, and fine, respectively. Grinding compound of numbers 400, 500 and 600 grit is intended for finish lapping, polishing and fitting of parts. Grit numbers 100 and 240 are intended for lapping and grinding chromium-cobalt hard-faced seats and disks.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title and mumber of this specification.
b. Type of vehicle and grit size (see 1.2).
c. Quantity.
d. Levels of packaging and packing (see 5.1 and 5.2).
e Additional packaging packing and markinf options, if desired.
f. If abrasive is other than specified (see 3.1.1).

MILITARY CUSTODIANS:

Army - MR
Navy - AS
Air Force - 84

Preparing activity:

Navy - AS

Review activities:

Navy - YD

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
SS-L-1682 LAPPING AND GRINDING COMPOUND, VALVE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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